

Fuzzy Logic and Fuzzy Sets

CSC3034 Computational Intelligence

Dr Tang Tiong Yew

Introduction

Crisp Set and Fuzzy Set

Fuzzy Logic vs Probability

Representation of Fuzzy Set

Linguistic Variables

Hedges

Operations of Fuzzy Sets

Introduction

Orange, Red, or Blue?



What is fuzzy logic?

- ✗ logic that is fuzzy
- ✓ logic to describe fuzziness
- ▶ based on the idea that all things admit of degrees
- ▶ for example,

The motor is running **really hot**.

Tim is a **very tall** guy.

Electric cars are **not very fast**.

High-performance drives require **very rapid** dynamics and **precise** regulation.

Sunway is **quite a short** distance from Puchong.

Tim is 181 cm, therefore he is a tall guy.

In **boolean or conventional logic**, a value is defined as the boundary to distinguish 'tall' and 'short'.

For example, the boundary is 180 cm. Samuel is 179 cm, therefore he is a short guy.

Is Samuel really a short guy, or an arbitrary line has been drawn?

Fuzzy logic makes it possible to avoid this situation.

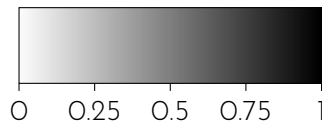
- ▶ Also known as multi-valued logic (unlike the classical logic with two values, true and false).
- ▶ Introduced by Jan Lukasiewicz, a Polish logician and philosopher in 1930s.
- ▶ He extended the range of truth values to all real numbers between 0 and 1.
- ▶ Lotfi Zadeh, a professor in UC Berkeley, published his famous paper 'Fuzzy sets' in 1965.
- ▶ Zadeh extended the work into a formal system of mathematical logic.

Boolean and fuzzy logic

Range of logical values



Boolean logic



Fuzzy/multi-valued logic

- ▶ Follows the principle of dichotomy.
- ▶ An element either belongs to a set or does not belong to a set.
- ▶ For example, if we have a set A and an element x ,
 x either belongs to the set A ($x \in A$),
or x does not belong to the set A ($x \notin A$).
- ▶ Characteristic function of crisp set A of universe X

$$f_A(x) : X \rightarrow 0, 1$$

where

$$f_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{if } x \notin A \end{cases}$$

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Monday	butter	coffee	Thursday	keyboard
Friday	laptop	cat	Saturday	Sunday

Set of NOT days of week	Set of days of week

Example 2

day of week	Set of weekends	Set of not weekends
Monday		
Tuesday		
Wednesday		
Thursday		
Friday		
Saturday		
Sunday		

- ▶ A set with fuzzy boundaries.
- ▶ Allows multi-valued logic.
- ▶ Fuzzy set A of universe X is defined by its membership function μ_A

$$\mu_A(x) : X \rightarrow [0, 1]$$

where

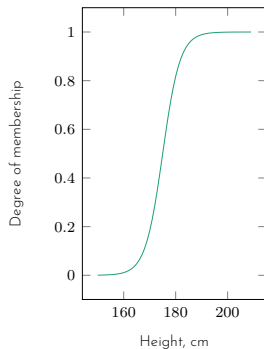
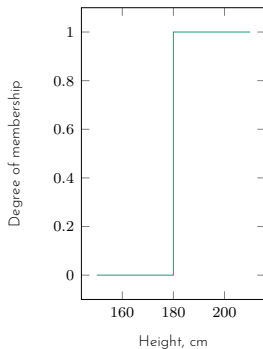
$$\begin{aligned}\mu_A(x) &= 1 \text{ if } x \text{ is totally in } A; \\ \mu_A(x) &= 0 \text{ if } x \text{ is not in } A; \\ 0 < \mu_A(x) < 1 &\text{ if } x \text{ is partly in } A.\end{aligned}$$

- ▶ The value of a membership function represents the degree of membership, or called the membership value of x in set A .
- ▶ More typically represents the human way of thinking.

Crisp set and fuzzy set

Set of tall men

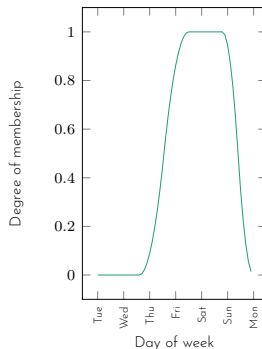
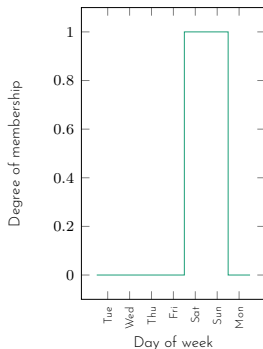
Name	Height (cm)	Degree of membership	
		Crisp	Fuzzy
Chris	208	1	1.00
Mark	205	1	1.00
John	198	1	0.98
Tom	181	1	0.82
David	179	0	0.78
Mike	172	0	0.24
Bob	167	0	0.15
Steven	158	0	0.06
Bill	155	0	0.01
Peter	152	0	0.00



Crisp set and fuzzy set

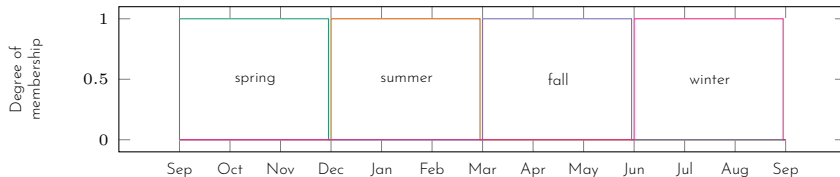
Set of weekends (truth values for 'weekend-ness')

	Weekends	
	Crisp	Fuzzy
Monday	0	0.00
Tuesday	0	0.00
Wednesday	0	0.00
Thursday	0	0.20
Friday	0	0.85
Saturday	1	1.00
Sunday	1	0.98

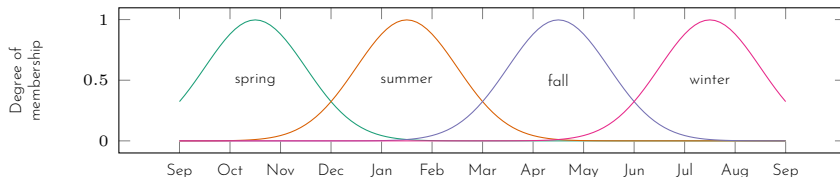


Crisp set and fuzzy set

Seasons of the year in southern hemisphere



Time of year



Time of year

Is crisp set or fuzzy set a better representation in this case?

Fuzzy logic $\neq$ Probability

Probability	Fuzzy Logic
represents the certainty of an event happening	not representing the (un)certainty
based on closed world model (everything is known)	not based on closed world model (there are things we do not know)
based on frequency and requires independence of variables	does not require statistical independence of variables
goes away once the event is observed	membership values persist after observation

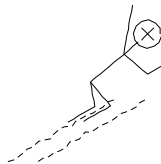
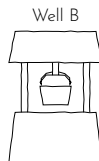
Fuzzy logic and probability

A man is lost in the desert. In the distance he can see two wells but he only has the strength to get to one.

Well A has a 0.9 probability of drinkable water

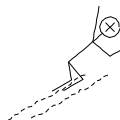
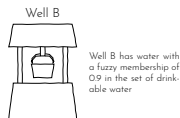


Well B has water with a fuzzy membership of 0.9 in the set of drinkable water



Which well should the man crawl to?

Fuzzy logic and probability



Which well should the man crawl to?

Well A

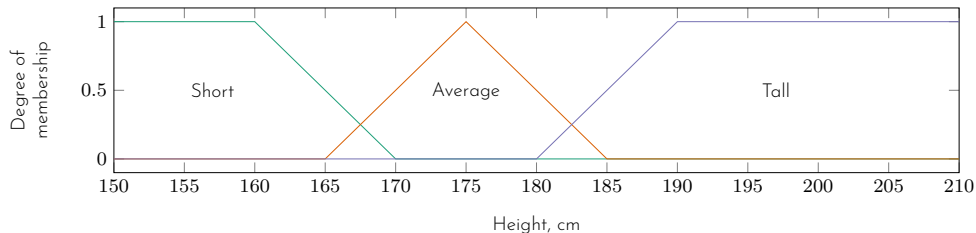
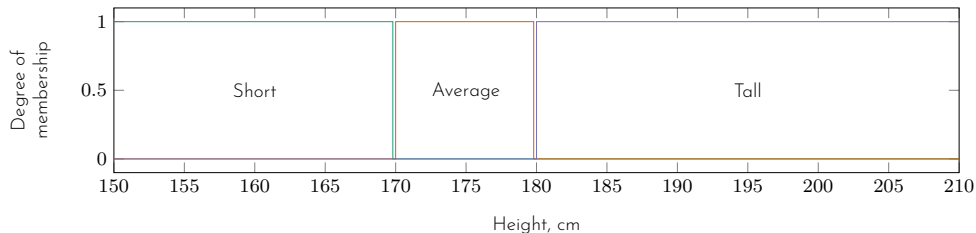
- ▶ The probabilistic information means that out of 10 wells, 9 will have suitable drinking water but 1 well will be poisonous.
- ▶ This one could be the poisonous one.

Well B

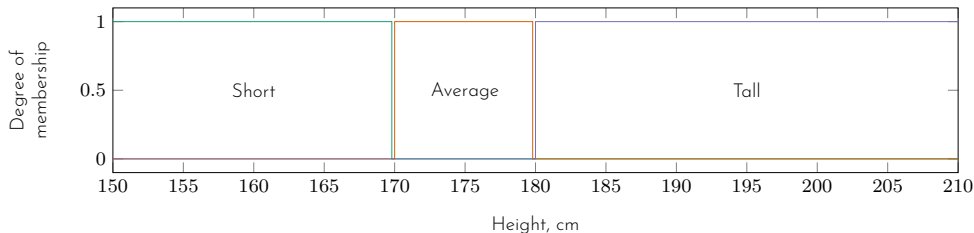
- ▶ The water quality is not perfect, but it is certain that the water is drinkable.

Representation of elements in a given set

Consider the crisp set and fuzzy set for the height of men.



Representation of elements in a given set



Consider the universe X of 5 elements,

$$X = \{155, 162, 172, 178, 182\}$$

How do we represent the crisp subset of short, average, and tall men in the universe X separately?

$$\text{short men} = \{(155, 1), (162, 1), (172, 0), (178, 0), (182, 0)\}$$

$$\text{average men} = \{(155, 0), (162, 0), (172, 1), (178, 1), (182, 0)\}$$

$$\text{tall men} = \{(155, 0), (162, 0), (172, 0), (178, 0), (182, 1)\}$$

Representation of elements in a given set

Consider the universe X with 5 elements,

$$X = \{x_1, x_2, x_3, x_4, x_5\}$$

If A is a crisp subset of X , and it consists of two elements,

$$A = \{x_2, x_3\}$$

A can be represented as

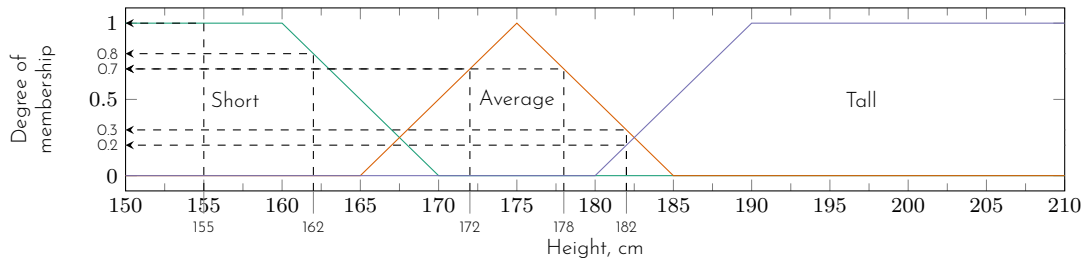
$$A = \{(x_1, 0), (x_2, 1), (x_3, 1), (x_4, 0), (x_5, 0)\}$$

i.e. in the form of $\{(x_i, \mu_A(x_i))\}$

where $\mu_A(x_i)$ is the membership function of element x_i in subset A .

Representation of elements in a given set

We can represent the fuzzy set in a similar format.



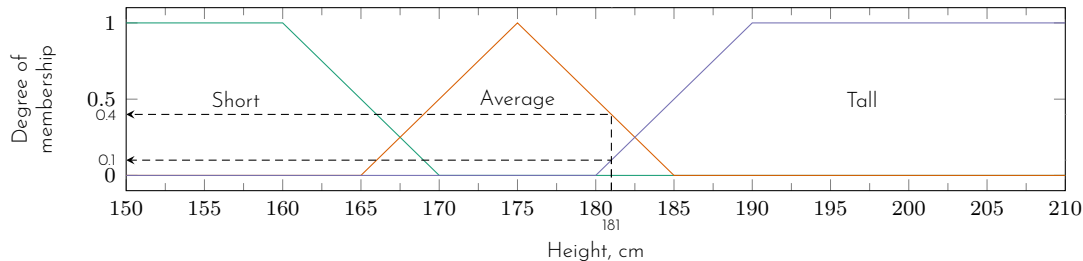
Consider the same universe $X = \{155, 162, 172, 178, 182\}$

The representation of the fuzzy subset of short, average, and tall men in the universe X is

$$\text{short men} = \{(155, 1.0), (162, 0.8), (172, 0), (178, 0), (182, 0)\}$$

$$\text{average men} = \{(155, 0), (162, 0), (172, 0.7), (178, 0.7), (182, 0.3)\}$$

$$\text{tall men} = \{(155, 0), (162, 0), (172, 0), (178, 0), (182, .2)\}$$



How do we determine the degree of membership for a man with the height of 181cm?

Let x represents the height, the degrees of membership of 181cm belongs to each subset are

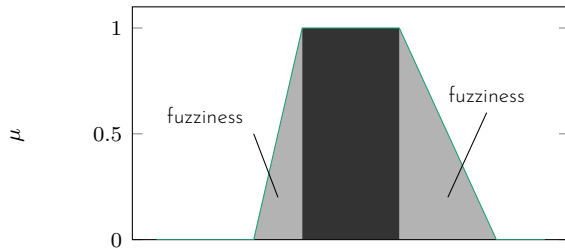
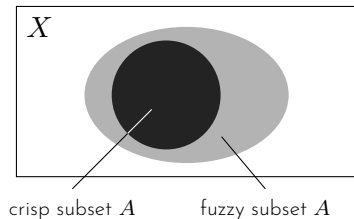
$$\text{short, } \mu_x = \text{short} = 0$$

$$\text{average, } \mu_x = \text{average} = 0.4$$

$$\text{tall, } \mu_x = \text{tall} = 0.1$$

Membership functions

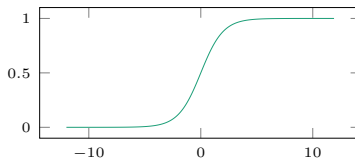
The difference between a crisp set and a fuzzy set is the introduction of fuzziness in membership function.



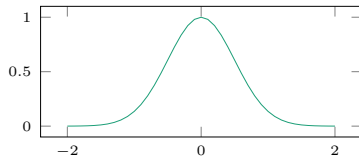
Membership functions

The typical membership functions used to represent a fuzzy set include

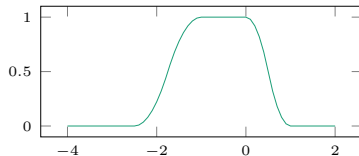
Sigmoid



Gaussian



Pi

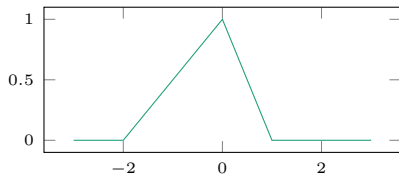


However, these functions increase the computation time.

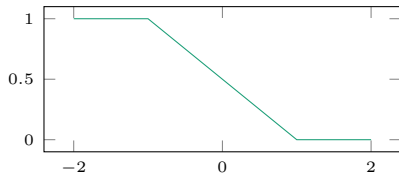
Membership functions

Therefore, in practice, most applications use **linear fit functions**

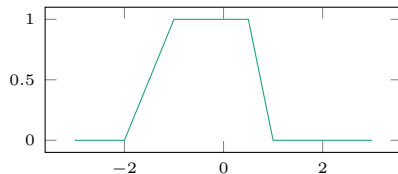
Triangular



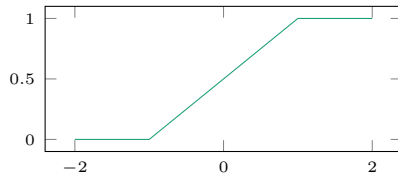
Z-curve



Trapezoidal

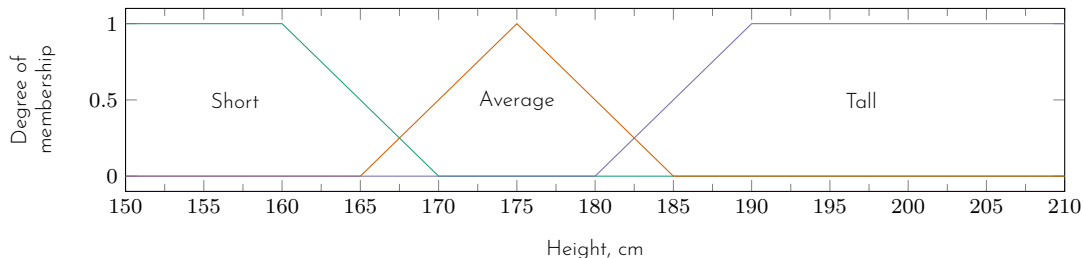


S-curve



Representation of a fuzzy set

A fit vector is used to described each fuzzy set. Using the example of the fuzzy sets of men's height,



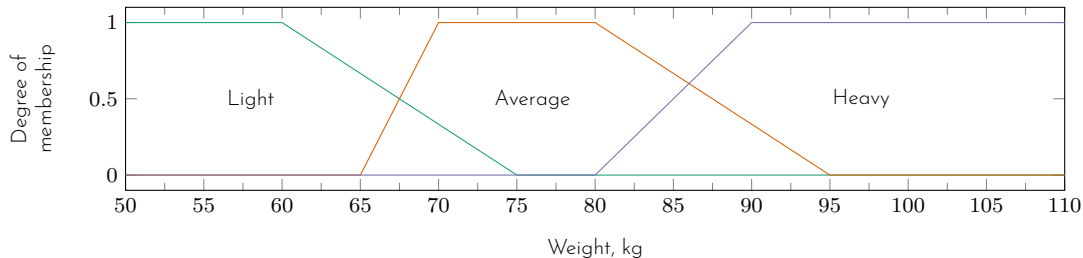
The fit vector for

short men = $(1/160, 0/170)$ or $(1/150, 1/160, 0/170)$

average men = $(0/165, 1/175, 0/185)$

tall men = $(0/180, 1/190)$ or $(0/180, 1/190, 1/210)$

Example 3



What is the fit vector for each fuzzy set of the fuzzy variable **weight**?

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- 'John is tall'

linguistic value: tall

'wind is strong'

linguistic value: strong

'speed is slow'

linguistic value: slow

- ▶ The range of possible values of a linguistic variable represents the universe of discourse of that variable.
- ▶ For example, 'speed'

universe of discourse: a range between 0 and 220km/h

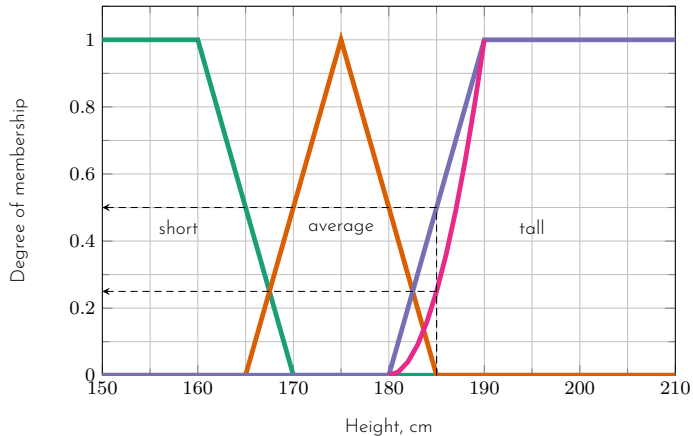
fuzzy subsets: very slow, slow, medium, fast, very fast

Each fuzzy subset represents a linguistic value of the corresponding linguistic variable.

- | | |
|-----------|--------------|
| a little | slightly |
| very | extremely |
| very very | more or less |
| somewhat | indeed |

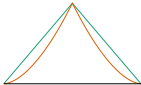
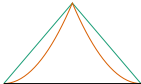
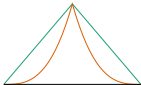
- ▶ Hedges act as operations.
- ▶ When a hedge is applied on a linguistic value, it changes the shape of the fuzzy set.
- ▶ For example, if 'very' is applied to a linguistic value A , it changes the shape from μ_A to $(\mu_A)^2$

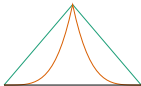
Example 4



'very tall' (i.e. applying 'very' on the fuzzy set 'tall') $\rightarrow \mu_A^{\text{very}} = (\mu_A)^2$

These are the hedges that are commonly used in practical applications:

Hedge	Mathematical expression	Graphical representation
a little	$(\mu_A)^{1.3}$	
slightly	$(\mu_A)^{1.7}$	
very	$(\mu_A)^2$	
extremely	$(\mu_A)^3$	

Hedge	Mathematical expression	Graphical representation
very very	$(\mu_A)^4$	
more or less	$\sqrt{\mu_A}$	
somewhat	$\sqrt{\mu_A}$	
indeed	$2(\mu_A)^2$ if $0 \leq \mu_A \leq 0.5$ $1 - 2(1 - \mu_A)^2$ if $0.5 < \mu_A \leq 1$	

Example 5

Using the fit vector for

$$\text{short} = (1/160, 0/170),$$

show the membership values of the descriptions

- ▶ very short,
- ▶ extremely short,
- ▶ very very short,
- ▶ slightly short,
- ▶ indeed short.

on a fuzzy set diagram.

Example 6

Given a fuzzy system of weights in kg with the fit vectors

$$\text{light} = (1/50, 0/60)$$

$$\text{average} = (0/55, 1/65, 1/70, 0/90)$$

$$\text{heavy} = (0/70, 1/90, 0/100)$$

$$\text{very heavy} = (0/95, 1/110)$$

Draw the fuzzy set diagram.

Show the membership values of 'more or less average' on the fuzzy set diagram.

Sets interact with one another through **operations**.

- ▶ Equality
- ▶ Complement (NOT)
- ▶ Containment (subset)
- ▶ Intersection (AND)
- ▶ Union (OR)

Equality of two sets

► Crisp sets

- Sets containing the same members are equal.
- $\{x_1, x_2, x_3\} = \{x_1, x_2, x_3\}$
- $\{x_1, x_2, x_3\} \neq \{x_1, x_2, x_4\}$

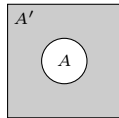
► Fuzzy sets

- Two sets are equal, if and only if, all elements have identical membership values.
- $\{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\} = \{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\}$
- $\{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\} \neq \{(x_1, 0.3), (x_2, 0.6), (x_3, 0.8)\}$

Complement (NOT)

► Crisp sets

- All the elements that are not in the set (A' or $\bar{A}$)
- For $X = \{x_1, x_2, x_3, x_4\}$ and $A = \{x_1, x_3\}$,
 $A' = \{x_2, x_4\}$

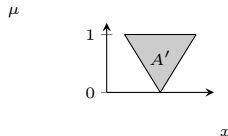
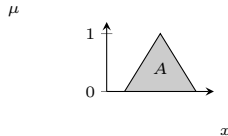


► Fuzzy sets

- Membership values of a complement set are given by

$$\mu_{A'} = 1 - \mu_A.$$

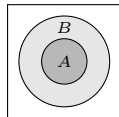
- If $A = \{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\}$,
then $A' = \{(x_1, 0.9), (x_2, 0.4), (x_3, 0.2)\}$



Containment (subset)

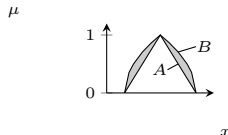
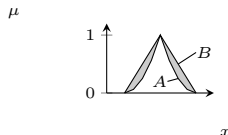
► Crisp sets

- $A \subset B$ if and only if all elements in A are also in B .



► Fuzzy sets

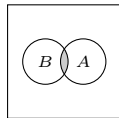
- $A \subset B$ if the membership values for each element in A is less than or equal to the membership value of the corresponding element in B .
- A hedge creates a subset or superset.
- If $A = \{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\}$ and $B = \{(x_1, 0.1), (x_2, 0.8), (x_3, 1.0)\}$, then $A \subset B$.



Intersection (AND)

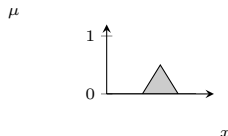
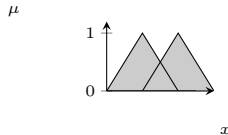
► Crisp sets

- The intersection of two sets contains those elements in both sets.
- If $A = \{x_1, x_2, x_3, x_4\}$ and $B = \{x_1, x_4, x_5, x_6\}$,
 $A \cap B = \{x_1, x_4\}$.



► Fuzzy sets

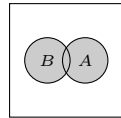
- The weakest element determines the degree of membership in the intersection.
- $\mu_{A \cap B} = \mu_A \cap \mu_B = \min(\mu_A, \mu_B)$
- If $A = \{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\}$
and $B = \{(x_1, 0.5), (x_2, 0.3), (x_3, 1.0)\}$,
then $A \cap B = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.8)\}$.



Union (OR)

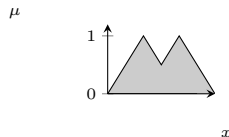
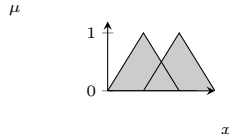
► Crisp sets

- All elements in either (or both) set(s) are included.
- If $A = \{x_1, x_2, x_3, x_4\}$ and $B = \{x_1, x_4, x_5, x_6\}$,
 $A \cup B = \{x_1, x_2, x_3, x_4, x_5, x_6\}$.



► Fuzzy sets

- The strongest element determines the degree of membership in the union.
- $\mu_{A \cup B} = \mu_A \cup \mu_B = \max(\mu_A, \mu_B)$
- If $A = \{(x_1, 0.1), (x_2, 0.6), (x_3, 0.8)\}$
and $B = \{(x_1, 0.5), (x_2, 0.3), (x_3, 1.0)\}$,
then $A \cup B = \{(x_1, 0.5), (x_2, 0.6), (x_3, 1.0)\}$.



Example 7

Given the universe of discourse,

$$X = \{x_1, x_2, x_3, x_4, x_5\}$$

and three fuzzy subsets,

$$A = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.8), (x_4, 0.6), (x_5, 1.0)\}$$

$$B = \{(x_1, 0.5), (x_2, 1.0), (x_3, 0.7), (x_4, 0.6), (x_5, 0.0)\}$$

$$C = \{(x_1, 0.3), (x_2, 0.1), (x_3, 0.2), (x_4, 0.8), (x_5, 0.4)\}$$

Find

- ▶ $A \cap B \cap C$
- ▶ $(A \cup B)'$
- ▶ $A \cap (B \cup C)$

CSC3034 Current Course Update

02 Fuzzy Logic and Fuzzy Sets - handout

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current Lab 2 uses scikit-fuzzy to build and evaluate Mamdani control systems.
- The physical-AI extension applies fuzzy brake/throttle logic to a simulated robot.
- The updated solution checks optional dependencies and supports a standalone fallback.

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See the synchronized lab notes and runnable examples for full instructions.